

Docker Sandboxes & Network Egress

Per-Agent Container Isolation, Port Sandboxing & Zero Data Leakage

Ironclad Isolation: One Container Per Active Worker

To ensure complete enterprise security and prevent cross-tenant data contamination, Fathom provisions an isolated Docker container for every active digital coworker via `crates/supervisor`.

CONTAINER SANDBOX SPECIFICATIONS

- **Capability Stripping:** All Linux capabilities dropped (`cap_drop: ["ALL"]`).
- **Privilege Escalation Blocked:** `no-new-privileges: true` enforced at Docker runtime.
- **Ephemeral & Persistent Volumes:** Dedicated volume mounts for browser cookies and workspace files.
- **Deterministic Port Mapping:** Dedicated loopback ports per agent (19000–19999 range).

STRICT NETWORK EGRESS GUARDING

The browser loopback service enforces strict firewall rules:

- **Cloud Metadata Blocked:** Rejects all calls to `169.254.169.254` (AWS/GCP metadata theft protection).
- **Private Subnet Deny:** Rejects connections to private RFC1918 subnets (`10.0.0.0/8`, `192.168.0.0/16`).
- **Multicast & Loopback Deny:** Prevents agents from scanning or probing internal corporate networks.



ENTERPRISE MULTI-TENANT SECURITY GUARANTEE

Even if an agent navigates to a malicious webpage attempting prompt injection or cross-site scripting, the attack remains completely trapped within the unprivileged Docker container with zero access to the host OS or other client workspaces.

Sovereign Data Security: Deploy multi-tenant AI operations with total confidence in cryptographic process separation and zero cross-client data leakage.